

Overlap Does Not Decide Independence

Unit 2 · Topic 2.7 · TODAY'S PROBLEM

Suppose a school surveys 200 students about whether they own at least one pet and whether they play a musical instrument. Let P be the event that a randomly selected surveyed student owns a pet and I the event that the student plays an instrument.

Aaliyah: *The events overlap because 84 students do both. Since they are not mutually exclusive, they must be independent.*

Marcus: *Overlap does not decide independence; the relevant probabilities must be compared.*

Pet ownership and instrument playing

Pet?	Plays	Does not	Total
Yes	84	36	120
No	16	64	80
Total	100	100	200

FIRST TAKE (5 min, on your sheet)

Does owning a pet seem to change the chance that a student plays an instrument? Give one table detail in one sentence.

- DOK 1** (a) State the counts for P , I , and $P \cap I$.
- DOK 2** (b) Find $P(I)$, $P(I | P)$, and $P(P \cup I)$. Show the fractions or setup.
- DOK 3** (c) Whose reasoning is supported? Decide independence using the two instrument-playing probabilities. Explain why the 84 students in both groups settle mutual exclusivity but do not settle independence. Write 3–4 sentences.

Video + follow-along: u4_lesson6_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

